

ABDUL WAHAB

MECHANICAL DESIGN ENGINEER AND PRODUCT DESIGNER

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Portfolio: <https://wahababdull.github.io/CV/>

Research Interests: Autonomous and intelligent vehicle systems, robotic navigation and SLAM, AI applications in robotics, scene understanding for autonomous driving, end-to-end autonomous systems, innovative product and mechanical design, and design for manufacturability in robotic platforms.

Mechanical Design Engineer and Product Designer with 2.0 years of experience specializing in SolidWorks-based CAD modeling. Skilled in 3D part design, complex assemblies, and detailed manufacturing drawings in compliance with engineering standards. Proven experience designing industrial machinery, mechanical systems, and consumer products, with a strong focus on accuracy, manufacturability, and functional performance. Adept at transforming concepts into production-ready designs through creative problem-solving techniques and industry standards. I am ready to help turn ideas into reality.

SKILLS AND EXPERTISE

- CAD and Manufacturing Design
- Design Simulation and Manufacturability Analysis
- Manufacturing Process Design and Design for Manufacturability (DFM)
- Engineering Problem-Solving and Root Cause Analysis
- Process Data Analysis

PROFESSIONAL EXPERIENCE

Mechanical Design Engineer | Freelance (Upwork)

August 2025 – Present

- Delivered professional **mechanical and product design services** to international clients via **Upwork**.
- Created 3D CAD models, assemblies, 2D technical drawings, and photorealistic renders for prototyping and manufacturing.
- Designed components for consumer products and industrial applications, optimized for material selection and manufacturing processes.
- Communicated directly with clients to gather requirements, manage revisions, and meet project deadlines
- Maintained high client satisfaction through **quality delivery and clear technical communication**.
- **Skills:** SolidWorks, 3D CAD Modeling, 3D CAD Rendering, 2D Technical Drawing, Clear Communication, Detail Oriented Design, Prototyping Support

Mechanical Design Engineer | Self- Employed | D12 Markaz, Islamabad, Pakistan

December 2024 – Present

- Designed precise 3D mechanical parts, assemblies, and production drawings using SolidWorks.
- Completed multiple mini and industrial projects, including Centrifugal Pumps, Heat Exchangers, Parabolic Troughs, and Phone Casings.
- Developed consumer and industrial products such as Mixture Machines, Helmets, Toothbrushes, Water Taps, and Waste Bins.
- Ensured manufacturability, dimensional accuracy, and functional performance in all designs.
- Collaborated with clients to convert ideas and concepts into practical, cost-effective design solutions.
- **Skills:** 3D Modeling, 3D Rendering, 2D Technical Drawing, Industrial & Product Design Consultation, Clear Communication, Detail Oriented

EDUCATION

Bachelor of Science | University of Engineering and Technology, Peshawar | Khyber Pakhtunkhwa, Pakistan

September 2019 – October 2023

- **Field:** Mechanical Engineering
- **Supervisor:** Professor Usman
- **Final Project:** Design and Fabrication of Mechanism for Automated Installation of 18 Liter water bottle of Water Dispenser

Faculty of Science | WENSAM college, Dera Ismail Khan | Khyber Pakhtunkhwa, Pakistan

September 2017 – June 2018

- **Field:** Pre-Engineering

PROJECTS

Design and Fabrication of Mechanism for Automated Installation of 18 Liter water bottle of Water Dispenser

January 2023 – March 2023

- It is our Final Year Project and I have designed its 3D model.
- Automated the manual process of lifting the water dispenser bottle, reducing human efforts.
- Building a customized solution to the problem, starting from lifting to adjustment on water dispenser, atomization
- Use of many mechanisms to optimize results.

3D model of ToothBrush using Solidworks

September 2021 – November 2021

- Modeling of parts: Create a detailed 3D design of the toothbrush
- Assembled: Assemble all component connections accurately
- **Problem:** Traditional toothbrushes contribute to waste and lack proper hygiene.
- **Solution:** CLI Toothbrush features a sustainable design with UV technology that eliminates 99.88% of bacteria and includes a travel-friendly case.
- **Benefits:** Reduces waste, ensures hygiene, and offers a compact, dual-purpose carry case for travel.
- **Functions:** Cleans teeth effectively, sanitizes with UV light, and provides portable protection.

3D model of Mixture Machine using Solidworks

September 2021 – November 2021

- Modeling of parts: Create a detailed 3D design of the mixture machine.
- Assembled: Assemble all component connections accurately
- **Problem:** Bulky, corded blenders; difficult cleaning; not ideal for single-serve use
- **Solution:** Compact, cordless, modular mixer with quick-assembly design
- **Benefits:** Portable, hygienic, space-saving, travel-friendly
- **Functions:** High-speed motor, stainless-steel blade, borosilicate glass jar, safety lock
- **Power:** 2000 mAh USB-C rechargeable battery

3D model of Mini Projects using Solidworks

September 2021 – November 2023

- Modeling of parts: Create a detailed 3D design and 3D rendering images
- Assembled: Assemble all component connections accurately
- **Problem:** Show versatile CAD skills across industrial, consumer, and conceptual designs.
- **Solution:** SolidWorks projects: heat exchanger, pump, solar trough, bottle cap, phone case, sculptural table, Mobius ring.
- **Benefits:** Versatile, manufacturing-aware, parametric, spanning industrial → consumer → abstract forms.
- **Functions:** Fluid power, thermal transfer, solar capture, sealing, device protection, structural support, kinetic demo.

LANGUAGES

- English (Professional)
- Seraiki (Native)
- Urdu & Hindi (Fluent)

CERTIFICATIONS

- Pakistan Engineering Council (PEC), Pakistan
- Certified SOLIDWORKS CAD Design Associate
- Lean Six Sigma White Belt and White Belt (LSSWB and LSSYB)
- Lean Fundamentals and Production
- Data Analytics in Lean Six Sigma
- CPR, AED and First Aid

REFERENCES

- **Professor Dr. Naveed Ahmad**
 - o Assistant Professor, UET, Peshawar, Khyber Pakhtunkhwa, Pakistan
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- **Professor Muhammad Alam Zaib Khan**
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